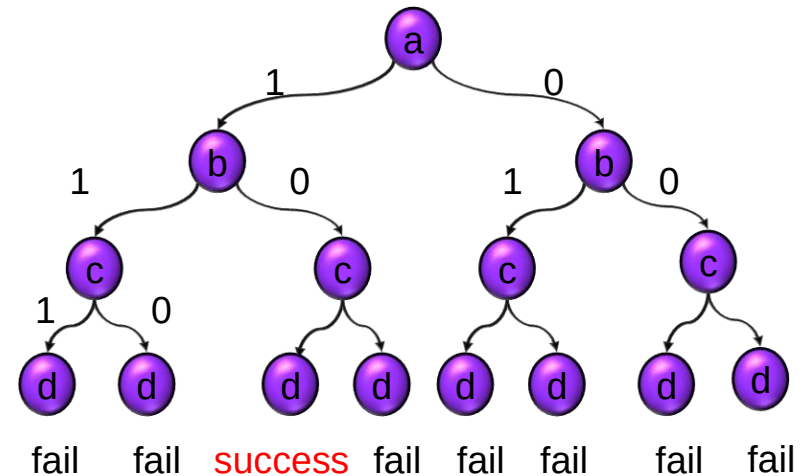


Combinational ATPG

- Introduction
- Deterministic Test Pattern Generation
 - ◆ Boolean difference*
 - ◆ Path sensitization**
 - ◆ D-Algorithm (1965)**
 - ◆ PODEM (1981)**
 - ◆ FAN(1985) **
 - ◆ SAT-based (1992)*
- Acceleration Techniques
- Concluding Remarks

*Boolean-based methods
**path-based methods



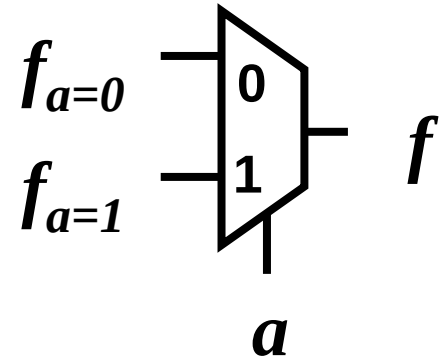
Decomposition

- Consider a circuit that realizes the function $f(a, b, c, \dots)$
- Fixed a to one:
 - ♦ **positive cofactor of f with respect to a**

$$f_{a=1} = f(a = 1, b, c, \dots)$$

- Fixed a to zero:
 - ♦ **negative cofactor of f with respect to a**

$$f_{a=0} = f(a = 0, b, c, \dots)$$

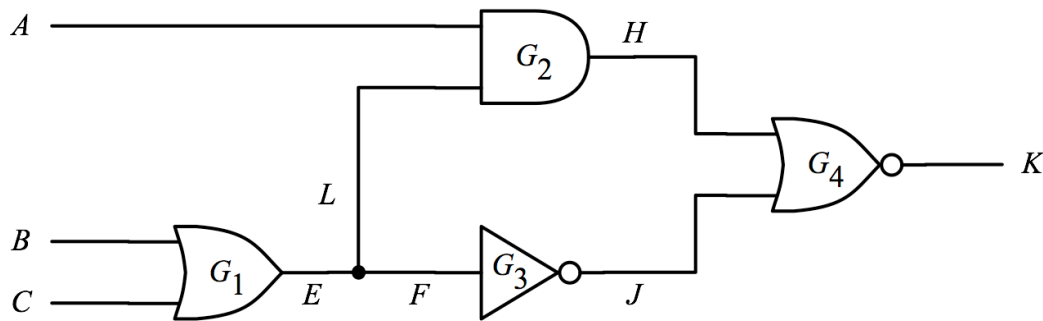


- **Shannon's Expansion** w.r.t. input a [Shannon 1948]

$$f = a f_{a=1} + a' f_{a=0}$$

Quiz

Q: Shannon's Expansion for output K w.r.t. input B
A:



Boolean Difference (1)

- **To detect a stuck-at zero fault**

- ♦ Good output (f) and faulty output ($f_{a=0}$) are different: $f \oplus f_{a=0} = 1$

- ♦ After Shannon Expansion: $[af_{a=1} + a' f_{a=0}] \oplus f_{a=0}$
 $= a[f_{a=1} \oplus f_{a=0}] + a'[f_{a=0} \oplus f_{a=0}]$
 $= a[f_{a=1} \oplus f_{a=0}] + 0$
 $= 1$

- ♦ Thus: $a[f_{a=0} \oplus f_{a=1}] = 1$

- **To detect a stuck-at one fault**

- ♦ Good output (f) and faulty output ($f_{a=1}$) are different: $f \oplus f_{a=1} = 1$

- ♦ After Shannon Expansion: $[af_{a=1} + a' f_{a=0}] \oplus f_{a=1} = 1$

- ♦ Thus: $a'[f_{a=0} \oplus f_{a=1}] = 1$

- **Boolean difference of f w.r.t. a**

$$\frac{df}{da} = [f_{a=0} \oplus f_{a=1}]$$

Boolean Difference (2)

- Boolean Difference = 1 means $f_{a=0}$ and $f_{a=1}$ are different
 - ♦ which means: a is sensitized to output f

$$\frac{df}{da} = [f_{a=0} \oplus f_{a=1}] = 1$$

- Example: AND gate, when $b=1$, $f = A$

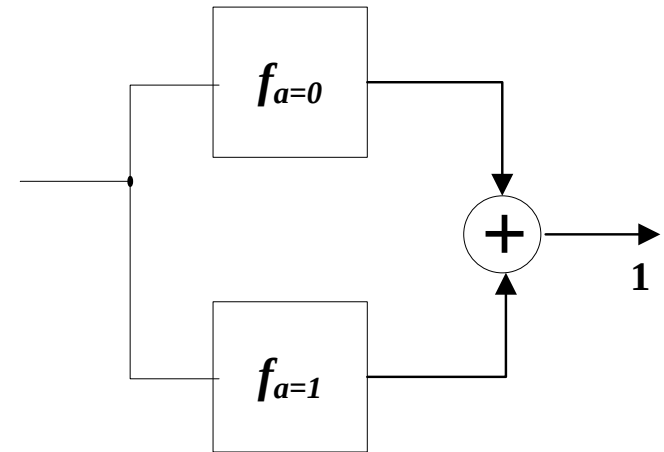
$$f = ab$$

$$\frac{df}{da} = [f_{a=0} \oplus f_{a=1}] = 0 \oplus b = b$$

- Example: OR gate, when $b=0$, $f = A$

$$f = a + b$$

$$\frac{df}{da} = [f_{a=0} \oplus f_{a=1}] = b \oplus 1 = b'$$



BD=1: sensitization condition

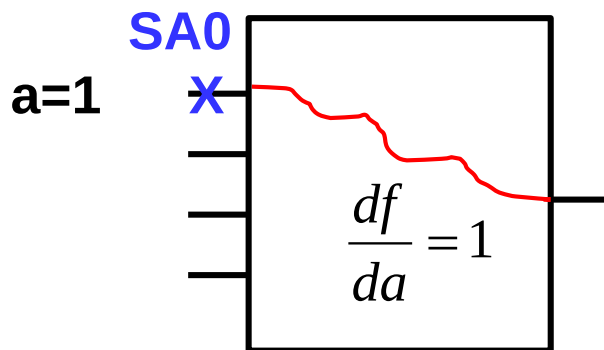
Boolean Difference (3)

- All patterns to detect a stuck-at-0 fault

- * $a = 1$: **fault excitation**

- * $BD = 1$: **sensitization** (aka **fault effect propagation**)

$$a \frac{df}{da} = 1$$



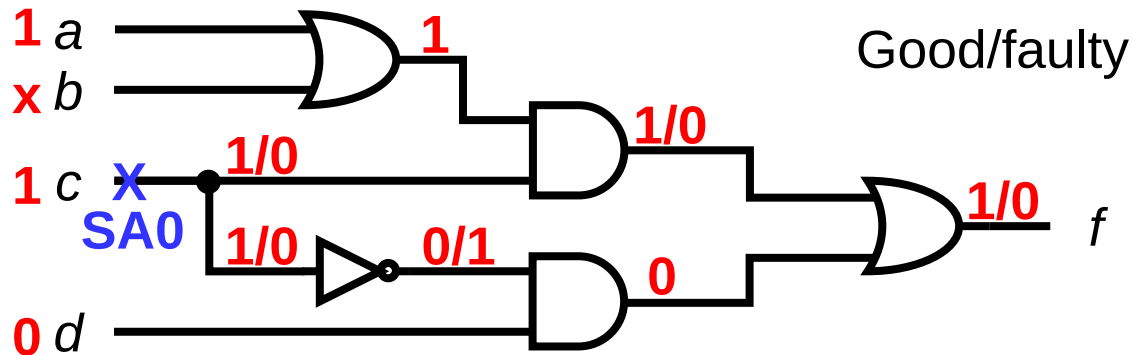
- All patterns to detect a stuck-at-1 fault

- * $a = 0$: **fault excitation**

- * $BD = 1$: **sensitization**

$$a' \frac{df}{da} = 1$$

Test Generation Example



- $f = (a + b)c + c'd$
- Set of all tests for *c stuck-at-0* is $c \frac{df}{dc} = 1$

$$\frac{df}{dc} = f(a, b, 0, d) \oplus f(a, b, 1, d) = d \oplus (a + b) = ad' + bd' + a'b'd$$

$$c \frac{df}{dc} = acd' + bcd' + a'b'cd$$

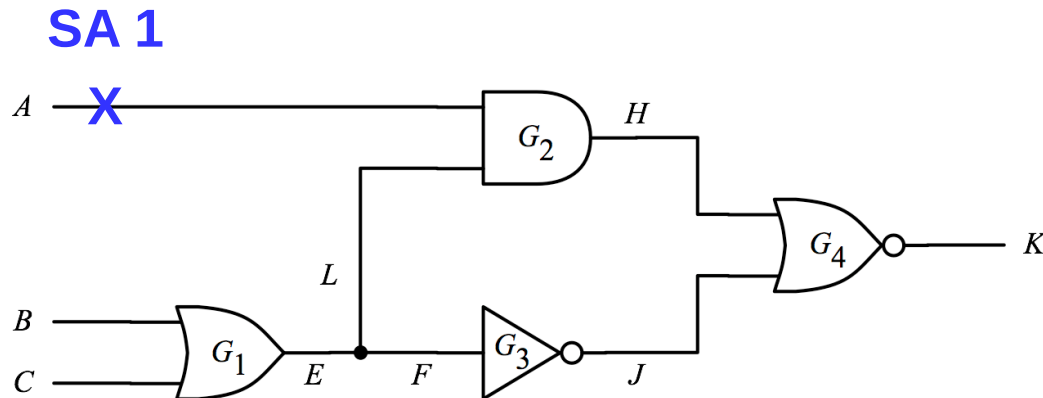
- Set of all tests = {1x10, x110, 0011} (x = don't care inputs)
- One *fully specified test pattern*: e.g. 1110
- One *partially specified test pattern (aka. test cube)*: 1x10

Quiz

(Cont'd) We already know $K = A'B'C + A'B$

Q1: Boolean difference $dK/dA = ?$

Q2: Use BD to find all test patterns for A stuck-at one fault.



Internal Faults

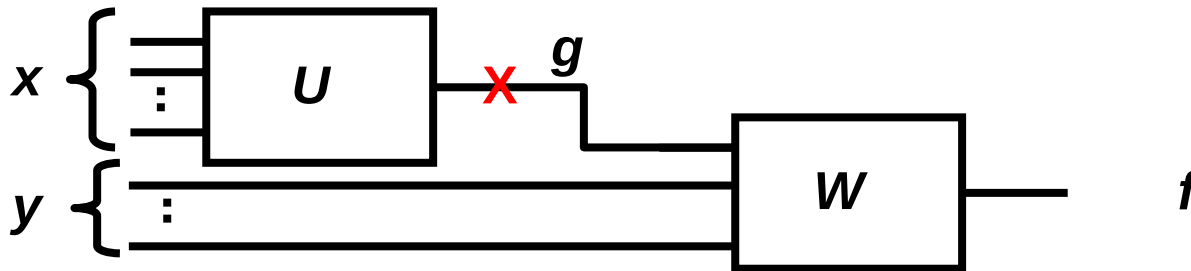
- Same approach also used for stuck-at faults internal faults
- Let g be internal signal of Boolean function f

$$f(x_1, x_2, \dots, y_1, y_2, \dots) = W(g, y_1, y_2, \dots)$$

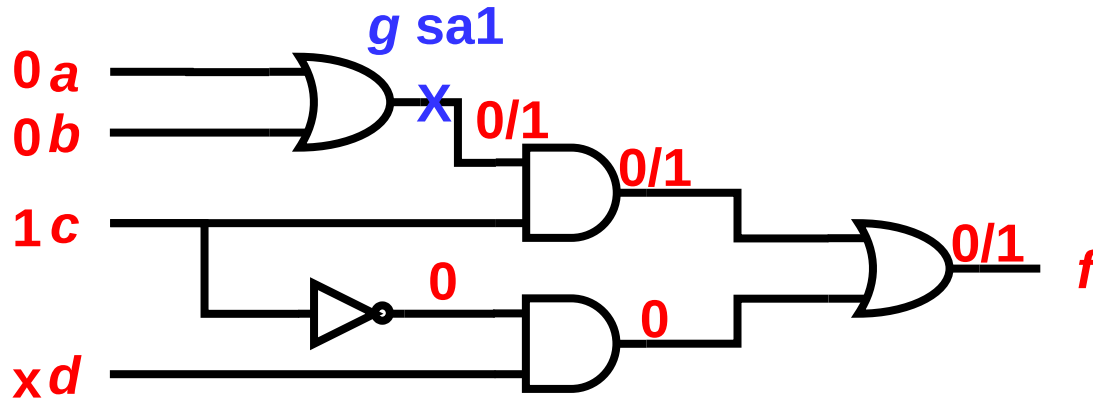
- Test sets for faults

- ♦ g SA0: $g \frac{dW}{dg} = 1$

- ♦ g SA1: $g' \frac{dW}{dg} = 1$



Example



$$f = (a + b)c + c'd \quad g = a + b$$

$$W = gc + c'd$$

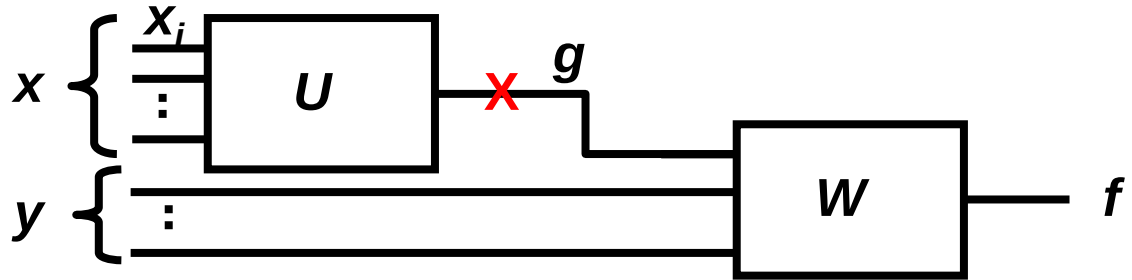
$$\frac{dW}{dg} = W_{g=0} \oplus W_{g=1} = cd' \oplus (c + c'd) = c$$

- Tests for g sa0
 $(a + b)c = ac + bc$
- Tests for g sa1
 $(a + b)'c = a'b'c$

Chain Rule

- Chain rule:

$$\frac{df}{dx_i} = \frac{df}{dg} \frac{dg}{dx_i}$$



- Example

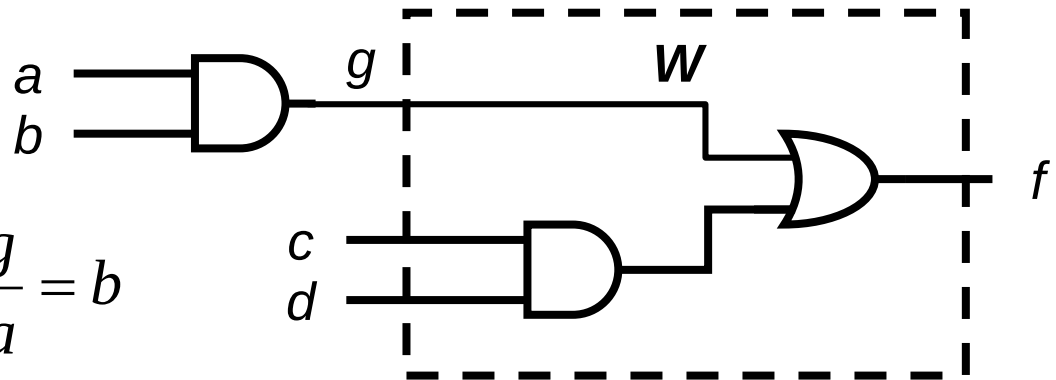
$$g = ab$$

$$W = g + cd$$

$$\frac{dW}{dg} = 1 \oplus cd = c' + d'$$

$$\frac{dg}{da} = b$$

$$\frac{df}{da} = \frac{dW}{dg} \frac{dg}{da} = (c' + d')b$$



Problem with Boolean Difference

- Boolean expression not always available
 - ♦ Especially in synthesized circuits
- Automatic algorithm is difficult
- Generating *ALL* test patterns for a fault is waste of time
 - ♦ Only need *ONE* test pattern for a fault

BD is Not Useful in Practice

Summary

- Deterministic Test Pattern Generation

- ◆ **BD=1** means sensitization condition

$$\frac{df}{da} = [f_{a=0} \oplus f_{a=1}] = 1$$

- ◆ Test patterns to detect a stuck-at-0 fault

$$a \frac{df}{da} = 1$$

- ◆ Test patterns to detect g stuck-at-0 fault

$$g \frac{dW}{dg} = 1$$

- ◆ Chain rule

$$\frac{df}{dx_i} = \frac{df}{dg} \frac{dg}{dx_i}$$

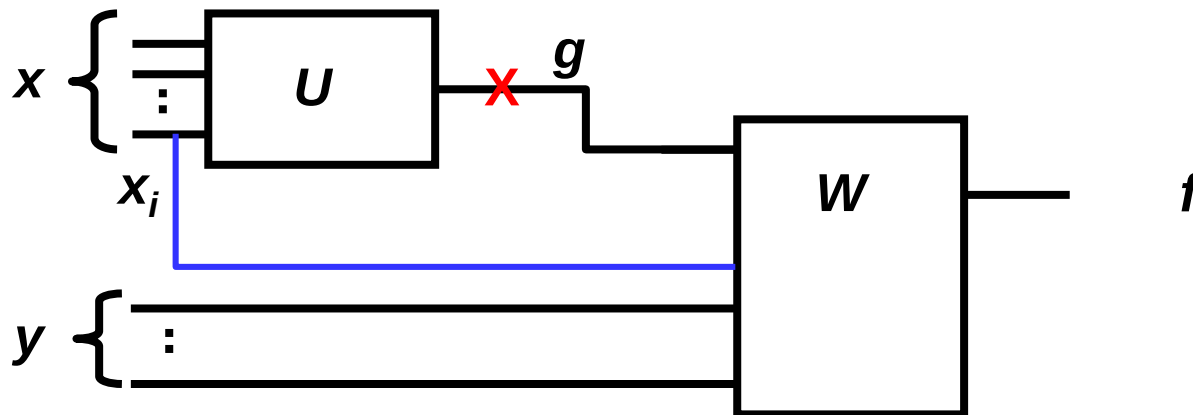
FFT1

- Q: Does it still apply if fanout exists?
 - ♦ U and W share some inputs x_i

- Test sets for faults

- ♦ g sa0: $g \frac{dW}{dg} = 1$

- ♦ g sa1: $g' \frac{dW}{dg} = 1$



FFT2

- Q: Does it still apply if fanout exists?

- ♦ U and W share some inputs x_i

- Chain rule:

$$\frac{df}{dx_i} = \frac{df}{dg} \frac{dg}{dx_i}$$

